Specific Aims

Background

According to National Survey on Drug Use and Health (NSDUH), 29.5 million people age 12 and older had alcohol use disorder (AUD) in the past year, whereas only 2.2 million (less than 10%) in this age group received alcohol treatment[1,2]. Provider availability in rural areas, fear of stigma, and economic hardship contribute to racial, gender and age disparities in treatment seeking[3–8]. The emergence of digital therapeutics (DTx) has provided new opportunities to address such health disparity in underprivileged groups with its higher accessibility, acceptability and at least modest efficacy[9–12]. The broad goal of this program of research is to improve the effectiveness of DTx, potentially by adding personal sensing technology and algorithms that allow DTx to have more refined capacities in anticipating lapses and recommending optimal treatment selection that meets individuals' momentary needs[13–15]

Machine learning models leveraging ecological momentary assessment (EMA) measures have performed relatively well to predict alcohol relapse risks, yet this methodology is associated with several limitations. As it actively relies on self-reports, it brings presumably large burden to users. This is particularly problematic as AUD is a chronic disease that requires constant risk monitoring. Reducing sampling rates and number of survey items might partially relieve this burden but at the cost of impairing prediction precision and temporal specificity. Further, this method is based on a top-down, theory-driven approach and are thus constrained to the scope of predictor selection and subject to limitations of previous findings. We might miss important predictors owing to gaps in our knowledge of alcohol relapse risk factors. Models built upon results from decades of research on White, male-dominant samples might also even exacerbate inequity when they are applied to DTx.

Passive sensing using text message monitoring represents new opportunities to address the above issues while still retaining some benefits of EMA. As text sensing collects data continuously and automatically, it imposes lower burden on users. Research has found that users report more willingness to use passive sensing measures for a longer period of time[16]. Some feature engineering techniques allow researchers to mine underlying constructs of texts. They are unconstrained and are thus less susceptible to gaps in theories and homogeneity in previous research samples. One benefit of collecting EMA measures is that the models are built upon robust predictors that are supported by decades of research. Some feature engineering methods can still seize benefits of this theory-driven approach by feeding algorithms with well-established risk factors. For example, sentiment analysis can reveal emotional tones, resembling subjective feelings measures in EMA.

Text messaging analysis can be informative of both *who* are at risks of relapses and *why* they relapse. Language analysis can have high predictability with regards to alcohol-related outcomes. Indeed, natural language processing (NLP) techniques on electronic health records and clinical notes have demonstrated moderate performance in predicting alcohol misuse, risky alcohol use and alcohol use status[17–21]. Further, although some NLP methods are hard to interpret, some might have potentials to generate meaningful findings that might optimize DTx design to customize recommended treatment at the moment. For example, LIWC (see *Section Customized Dictionary*) might deepen our understanding of psycholinguistic risk factors[22] and sentimental analysis weighing emotional tone might have implications for emotion-focused therapy. Topic modeling that examines hidden semantic patterns and identifies clusters of topics might reveal cautionary topics that trigger alcohol use.

Current Study

In the current study, I will use different feature engineering techniques combined with various statistical algorithms to analyze participants' SMS messages aiming to predict their relapse risks. Broadly, I will evaluate both the predictability and interpretability of each distinct method. Followings are the more specific aims:

Aim 1: Train and evaluate predictive performance of machine learning models built upon language features derived from different NLP techniques to predict alcohol relapse risks. We will conduct various feature engineering methods such as topic modeling, sentiment analysis, bag of words, word embeddings and customized dictionary. For each distinct feature set derived from these natural language processing techniques, we will build machine learning models including glmnet (penalized, parametric linear model), k-nearest neighbor (KNN), linear/quadratic discriminant analysis (LDA/QDA), random forest (RF), and neural network (NN).Predictability of alcohol lapse risks will be evaluated based on prediction accuracy.

Aim 2: Identify important features and evaluate their interpretability with respect to recommending interventions. Model interpretation is key to providing treatment recommendations and uncovering potential relapse causes. We will calculate both global feature importance and local feature importance across all feature engineering techniques and machine learning algorithms. We will qualitatively assess top-performing features (i.e., with high feature importance).

Aim 3: Examine model fairness in historically underprivileged subgroup populations. It is also important to note that if embedded algorithms perform relatively worse for marginalized groups, their use can exacerbate rather than alleviate treatment disparities. As such, model performance between privileged vs. unprivileged groups should be carefully examined. We will evaluate model performance in demographic subgroups that face excessive barriers accessing alcohol treatments or medications, including females, racial minorities, individuals living under poverty, and older population. The performance metrics we choose are area under the receiver operating characteristic (auROC), area under the precision-recall curve (auPR), positive predictive value (PPV), true positive rate (sensitivity), and true negative rate (specificity).

Significance

AUD remains a prevalent, lasting and costly problem in the United States over the past few years. According to 2022 NSDUH[1,2], an estimate of 29.5 million (10.5% of the population) individuals aged 12 or older had AUD in 2022, which is consistent with the estimates for 2021. The percentage of adults who had AUD was steady between 2011 and 2014, with slight decrease after 2014[23]. Nonetheless, AUD-related mortality and alcohol use frequency has spiked during the COVID-19 pandemic[24,25]. AUD problems have also brought huge economic burdens to the United Stataes, where excessive alcohol use has cost U.S. $223.5 billion in 2006 and $249 billion in 2010[26].

Despite the high prevalence of AUD, few are able to access alcohol treatments due to various barriers. Only a small portion (7.6%) of individuals aged 12 or older with AUD received alcohol use treatment in the past year[1,2]. A systematic review study has identified three major sources of barriers: shame and stigma, lack of awareness in treatment need, and fear of giving up drinking[27]. Several studies have identified that AUD patients face financial barriers[6,7], lack of knowledge or awareness[28,29], social stigma[30–32], and geographical barriers[33] for accessing care. Individuals with disadvantaged demographic background suffer from even heightened barriers to treatment resources. Women with high severity of alcohol use face larger fear of stigma compared to their male counterparts[31]. Disparities in treatment-seeking behavior based on race, gender, and age are also evident[3–8].

DTx is an emerging tool that shows promising potentials in addressing extant challenges AUD patients face in accessing alcohol treatments. DTx are evidence-based health software that generate or deliver medical intervention to patients at the goal of treating, alleviating, or managing a disease or disorder. They show effectiveness in treating mental health conditions and demonstrate clear clinical advantages[34,35]. Further, DTx show significant potentials in reaching hard-to-reach populations, including those from socially marginalized group who encounter increased barriers to access treatment. They can benefit patients from rural areas with low provider availability as they are accessible remotely via mobile phones, tablets, or other devices[10,36]. DTx also show potentials to relieve barriers to seek professional help resulting from stigma concerns as they can facilitate anonymity and shorten the need for in-person interactions. They are of lower costs compared to in person care and have the capacity to be scalable.

Benefits of DTx can be further extended to achieve risk monitoring and treatment recommendation through incorporating personal sensing technology and machine learning algorithms. With implemented algorithms for relapse prevention, DTx can identify *who* are at heightened risks of lapsing, *when* they will lapse, and *why* they are at increased risks. Notably, these functions are very important given the dynamic nature of alcohol relapse. AUD is a chronic disease where patients face constant challenges of relapsing after abstinence[37,38]. Indeed, current work has expanded the relapse prevention model to reflect the temporal, multifaceted, and dynamic process of alcohol relapse[39]. The model now categorizes alcohol relapse precursors into two classes – distal factors and proximal factors[39–41]. Distal factors refers to stable individual traits such as socioeconomic status and education background. Proximal factors (e.g., drinking partner, drinking behavior) are situated and can be fluctuating over time. Proximal factors usually occurs before an episode of alcohol use[42]. They can be elusive to therapists' attention due to infrequent visits, highlighting importance of sustained monitoring to prevent lapses.

One critical question when building such machine learning algorithms is to pick which outcome to predict. A universally accepted formal operational definition of alcohol relapse has not yet been established due to the multidimensional nature of relapses[43]. Relapse usually refers to the return of a symptomatic behavior. Current perspectives on conceptualization of alcohol relapse can encompass: 1) the level of alcohol consumed; 2) time of abstinence or reduced use; and 3) negative consequences related to use. In this study, our primary outcome is focused on lapse, defined as a single episode of alcohol use. Lapses can serve as an early warning sign of failure to sustain a desired behavioral change[44,45]. Research has shown that initial lapse and frequent lapses are the associated with enhanced risks of relapses[46,47].

Another crucial question in lapse risk prediction is determining which predictors to input into the algorithms. One potential method relies on a top-down approach that involves integrating domain-specific knowledge from a relapse prevention model into the algorithms. Alcohol relapse risk factors are multifaceted and might involve

affect state, social interactions, and alcohol-related factors. For instance, the majority of an inpatient teen sample reported initial relapse to alcohol when offered alcohol, when in negative state, and when in interpersonal conflicts[48]. Other commonly found risk factors include alcohol craving[49,50], negative affect state[49], cognitive factors[49], and interpersonal problem[49]. Alternatively, researchers could adopt a bottom-up approach without pre-specified indexes when building models. It starts with collecting data from various sources and then analyzes them at a granular level and extracts relevant features. This method is less susceptible to our knowledge gaps and might even discover previously unfound underlying processes contributing to alcohol lapses. We might even expect less differential performance under the bottom up approach across privileged vs. unprivileged groups because the features are not drawn from past research dominated by White males with AUD and are thus potentially unbiased towards privileged groups.

Importantly, one of the key benefits of DTx is that they partially address barriers accessing professional help by providing 24/7/365, affordable, personalized support. However, they might exacerbate health inequity if embedded algorithms perform relatively worse for less privileged groups. One possible explanation of differential performance across subgroups might stem from the under-representation of unprivileged groups in the dataset. It is therefore important to recruit a diverse sample in the training dataset. Another reason for divergent model performance between subgroups might be owing to the predictors we feed the algorithms. Unbiased predictors should be carefully selected during model training.

Recent DTx work has found that EMA measures can yield satisfactory performance in predicting lapses. Our group developed an XGBoost machine learning model built upon self-reported craving, affect, efficacy, risky situations, stressful events, pleasant events to predict alcohol lapses in the next hour, day, or week[51]. The surveys were collected up to four times daily for three months. The model achieved exceptional performance when predicting lapses for new individuals, with a mean auROC score of .89, .90 and .93 for the hour-, day-, and week-level model respectively. We also further examined global feature importance and discovered that past alcohol use and future self-efficacy consistently contributed greatly across all models to predict lapses. Examining local feature importance in these models might is also capable of identifying the factors that contribute to lapse risk for any specific person and moment in time.

Notheless, relying on EMA measures for model building is associated with several limitations. First, the burden of constantly completing surveys makes it sustainably infeasible for real-world DTx use. Most EMA relevant mental health research time windows last from two weeks to three months[52–56], which is insufficient for real-world DTx use. As extended period of time of app use is anticipated, users' perceived burden of answering surveys is presumably larger[57]. Although minimizing the number of items in the surveys and the frequency of prompting users to complete the surveys might help mitigate the associated burden, it can inevitably reduce the prediction precision and temporal precision of algorithms.

Second, decisions regarding what constructs to assess and what items to include to assess these constructs are limited by theory and past data. Including risk factors solely drawn from decades of research on White, male-dominant samples might even exacerbate health disparities when applied to DTx. Unfortunately, our current understanding of alcohol relapse precursors is not comprehensive, especially for proximal factors. For example, Marlatt's proposed taxonomy characterizes high-risk situational precursors to alcohol relapse such as social pressure and positive/negative emotional state[58]. Nonetheless, replication studies have found this theoretical framework to be somewhat unreliable and have low predictive validity of post-treatment outcomes[59–61]. Further, while there is a growing recognition of the intricate interplay between distal and proximal factors in the relapse process, more evidence is needed to support this theoretical construct. A review study has also identified methodological constraints in aggregated alcohol research, highlight that there was a dearth of "near real-time" data capturing proximal factors in alcohol relapses[40].

Text sensing technology, which is both feasible and sustainable, represents new opportunities in DTx that might address limitations of the current active reporting approach. Since AUD is a chronic condition requiring ongoing risk monitoring over an extended period, DTx can benefit from SMS sensing because it places a low burden on

users and allows for continuous data collection. EMA studies collecting passive data have demonstrated high acceptability from participants and higher compliance rates compared to active measures[9,16]. Risk monitoring using SMS sensing is temporally sensitive to fluctuating risks as it can detect potential triggers in messages in time without actively prompting users to reflect on their feelings at the moment or report their environment.

With the burgeoning of NLP techniques, language analysis is well-equipped with diverse feature engineering tools to accomplish multiple tasks including prediction of alcohol-related outcomes in new individuals. Some might have good performance because they allow researchers to mine previously established robust construct of alcohol lapse precursors from SMS messages and incorporate them in training algorithms. For example, a bag-of-words approach using customized dictionary might allow researchers to mine alcohol-related content from messages such as alcohol cravings. Sentiment analysis can also enable researchers to assess individuals' affect state. Further, some NLP methods follow a bottom-up approach and are therefore less susceptible to limitations in past research. For example, topic modeling techniques such as Latent Dirichlet Analysis (LDA) and Latent Semantic Analysis (LSA) can examine patterns in SMS messages and identify clusters of topics from the data. Word embeddings approach transforms words into vectors with high dimension space.

NLP also offers avenues for model interpretation, which is critical in DTx applications as it may yield valuable insights into treatment recommendations. Although some feature engineering techniques (e.g., word embeddings) might reveal arbitrary features that is incomprehensible, some can still mine features that are interpretable. Assessing feature importance helps us understand how features contribute to the models (i.e., which are robust in predicting lapses). For example, global feature importance can identify features robust across individuals in predicting lapses. Local feature importance provides insights on what contributes to a lapse for a specific person at a specific time.

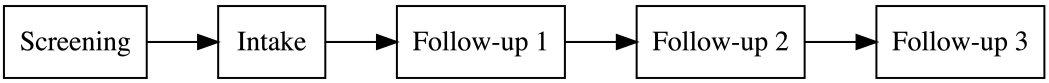
Approach

Overview

This study will analyze a subset of data from a larger grant funded by National Institute of Alcohol Abuse and Alcoholism (R01 AA024391). It is important to note that additional measures were collected from the participants that are not directly related to this study.

Participants

Individuals at early stages of recovery from AUD were recruited from Madison and surrounding area via social media platforms (e.g., Facebook), referrals from clinics, and television and radio advertisements. After initial phone screen, interested individuals came in-person to complete a more in-depth screening to determine their eligibility. We documented their demographic information. Inclusion criteria include that participants: 1) must be at least aged 18 or older; 2) must meet criteria for AUD with at least moderate severity (>four DSM-5 criteria); 3) must be abstinent from alcohol for at least one week and fewer than two months at time of intake; 4) must be able to read and write in English; 5) must be willing to use smartphone and their smartphone is compatible with our study technology. Participants were excluded if they have a lifetime history of severe and persistent mental illness. Of the 216 interested participants, 199 met eligibility and enrolled in the study. [final sample size for text analysis?] ## Procedures The study lasted up to three months with five in-person visits (see *Figure 1*). Participants completed a semi-structured interview at intake and each of the three one-month apart follow-up visits about their frequently communicated contacts. Interview questions include contextual information about each contact person (e.g., past-drinking history with this individual) and risk-related information (e.g., pleasantness or unpleasantness of interacting with this individual). At intake, participants reported contact information retrospectively. At each follow-up visit, the researchers directly backed up participants' text messages and asked contact information on newly-identified people. Participants were instructed on how to delete text messages that they did not want to share with the resesarch team. Apart from these in-person visits, participants also reported their alcohol use when completing daily surveys prompted up to four times per day. These surveys also included measures of alcohol craving, risky situations, stressful events, and pleasant events. We further passively collected participants' geolocation data and asked contextual information about their frequently visited places (not relevant to this study).



Source: Article Notebook

Figure 1: Flowchart of in-person visits. We obtained participant demographics and alcohol use history from the screening session. Measures related to frequent contacts were reported at intake and the three follow-up visits. SMS messages were downloaded from participant phones at each of the follow-up visit.

Measures

Individual Characteristics

We collected participants' individual characteristics including their demographics and their past drinking history during the screening session (see *Table 1*). It is important to note that those measures will be included as predictors of relapse risks in training machine learning algorithms in this study. A subset of these measures might also be used to evaluate model fairness on subgroups (e.g., sex, race, income, etc.).

SMS Messages

At each of the follow-up visit, a research assistant backed up the participants' SMS messages from their phone. Measures included the message type (incoming vs. outgoing), date and time sent/received, text body, contact name, whether the participants read the text or not, etc. Images and voice texts were excluded from analysis. Both group messages and one-on-one messages were obtained from participants' phones. We will only analyze participants' individual outgoing messages in this study. ### Contact Information Participants self-reported their frequent contacts at intake. Contextual information about the contact person was collected (see *Table 1*). Throughout the course of the study, we also documented any additional frequent contacts if the participant interacted with them at least twice in the past month. Same contextual information was obtained.

Table 1: Participant self-reported measures

Log Type	Measure
Demographics	Age
	Sex
	Race
	Ethnicity
	Highest Education
	Employment Status
	Total Personal Gross Income
	Marital Status
	Family Member
Alcohol Use	Alcohol Use History
	DSM-5 Checklist for AUD
	Young Adult Alcohol Problems Test
	WHO-The Alcohol, Smoking and Substance Involvement Screening Test
Contact Information	Relationship of Contact to Participant
	Drinking History with Contact
	Drinking Status
	Whether Contact Would be Present in Future Drinking
	Whether Contact is in Recovery
	Support from Contact
	Emotional Experience with Contact

Alcohol Lapses

Participants were prompted up to four times daily to report their recent alcohol use. Dates and times of any unreported past alcohol use were obtained in these surveys.

Feature Engineering

Text messages will be the sole predictor in the analysis. We will use the following natural language processing techniques to transform SMS messages into machine-interpretable data: 1) topic modeling; 2) sentiment analysis; 3) tf-idf; 4) word embeddings; and 5) customized dictionary. Each feature engineering technique will be evaluated separately. The outcome variable is dichotomous (Lapse vs. No Lapse). We will predict alcohol lapses in the next 24-hour window (i.e., next day lapse prediction). Predictions are updated every hour so that all algorithms will

provide hour-by-hour estimation. The first prediction window will start from the midnight of the second day of participation and end 24 hours later.

Topic Modeling

Topic modeling is an unsupervised machine learning method to identify clusters of topics in a body of text. It can do text mining without having predefined labels. In this study, I will use one of the most popular topic modeling technique – Latent Dirichlet Analysis (LDA). LDA unravels the hidden structure in the relationships between words in a document and groups similar words into topics. It assumes that different documents (i.e., SMS messages) have same topic clusters but with different distribution. As such, different topic clusters might have common words.

Sentiment Analysis

Research has found that positive/negative emotional state to be associated with increased alcohol relapse risks[40,48,49]. Sentiment Analysis infers the emotional underpinning of texts. I will use the *nrc* package available in R to mine emotional tone in SMS messages[62]. Nrc is a built-in dictionary that assigns a binary label (yes/no) to each lexicon (i.e., single word) into categories of positive and negative sentiment, as well as the eight basic emotions – anger, fear, anticipation, trust, surprise, sadness, joy, and disgust. The annotation task was accomplished via crowd sourcing. We will assign each SMS message a score in those ten categories by summing sentiment content of all words in that message.

Tf-idf

In this study, I will leverage combination of both term frequency (tf) and inverse document frequency (idf) to indicate word importance inside a document (see below formula). This scoring method has the advantage of highlighting the importance of a word to a document in a collection of documents. It has important implications in analyzing SMS messages so that we have a better understanding of the robustness of a certain word to a single message across all text messages that the participants sent. The idf score minimizes weights of words that are commonly used across all messages and can highlight words that contain important information.

$$\text{tf} = \frac{n_{terms\ in\ document}}{n_{terms}}$$

$$\text{idf} = \ln\left(\frac{n_{documents}}{n_{documents\ containing\ term}}\right)$$

$$\text{tf-idf} = tf * idf$$

Word Embeddings

Word embeddings are vectors that represent relationships of words within a document in a lower-dimension. I will use the pre-trained Bidirectional Encoder Representations from Transformers (BERT)[63] to transform SMS messages into such machine-readable vectors. They are capable of capturing embedded semantic structure within a neural network architecture. They take into account the context of where a word is (i.e., word tokens before and after the target word).

Customized Dictionary

A bag-of-words approach represents word occurrences inside a document. This study will expand this total word count approach by using customized dictionaries built upon the LIWC dictionary[64] and compute score by counting frequency of words that belong to each category. LIWC aligns with the current alcohol relapse risk factor literature in that it examines psychometric properties including cognitive state and social processes[48,49,49]. In addition to these linguistic categories, we will also include customized alcohol-related category as predictors as previous research has found them to be robust indicators of alcohol lapses[49,50].

Candidate Machine Learning Algorithms and Model Training

Candidate algorithms that will be used include glmnet (penalized, parametric linear model), k-nearest neighbor (KNN), linear/quadratic discriminant analysis (LDA/QDA), random forest (RF), and neural network (NN). A random split of 3/4 of the dataset will be used for model selection and model training. For each algorithm, we will perform a grouped, nested cross-validation on the training set to select the best model configuration. The selected best model will then be trained on the whole training dataset.

Model Performance Evaluation and Feature Importance

The primary performance metrics to select the best model configuration and evaluate the model performance on the test sets will be auROC. auROC computes the area under the ROC curve which demonstrates the trade-off between sensitivity and specificity across all possible thresholds. We choose this metric because it is more informative than accuracy for imbalanced data. Values between .70 and .80 are considered fair, values between .80 and .90 are considered good, and values above .90 are considered excellent.

Predicted probabilities will be calculated for all test sets. Binary classification (Lapse vs. No Lapse) will be determined with optimal decision threshold determined by Youden's Index.

We will also calculate SHAP values to interpret the results. SHAP is a game theory based method to explain how each feature influences the model output. It assigns importance to each feature, where a positive feature importance positively affects the model output. This methodology can be applied to any machine learning algorithm and can increase model transparency and interpretability.

Algorithmic Bias Evaluation

In addition to auROC, we will also evaluate auPRC, sensitivity, specificity and PPV across different subgroups. Negative Predictive Value (NPV) is not included due to the highly imbalanced outcomes in our dataset ($N_{positive}$ = , $N_{negative}$ =). We will compare these performance metrics between different sex, race, income and age subgroups as these demographic minority groups face enhanced barriers to access alcohol treatment due to stigma, financial difficulties, etc. They are also traditionally underrepresented in research settings.

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